Yaswanth Muppalla

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AI Platform Engineer | ***Senior Software Engineer*** | *Backend & Agent Systems* | ***AWS/GCP***

PROFESSIONAL EXPERIENCE

American Express | Senior Software Engineer | AI Engineer **Apr 2023 - Present**

Impact: Delivered **AWS-based Java/Python APIs** and **IRIS AI platform services** plus **Kafka/Flink/Airflow pipelines for digital banking**, improving reporting efficiency by **35%**, achieving **99% data accuracy**, reducing rerun/debug cycles by **3x**, and accelerating applied AI delivery for **credit-oversight copilots**.

- Partnered with BCA/CCA data pipeline teams and built **JSON-configurable Apache Flink pipelines on-premises** for ingestion, transformation, joins, validation, and downstream publishing, later migrating them to **GCP** for digital-banking and credit-oversight use cases.
- Built **Airflow workflow automation** with dependency controls, rerun handling, SLA monitoring, data-quality checks, and failure alerting; published governed outputs to **BigQuery** and exposed controlled analytical paths through **AWS API Gateway, Lambda, and ECS** for reporting applications and digital-banking products, with **stable data contracts, retries, and access controls**.
- Designed an **IRIS-backed RAG ingestion platform** with document extraction, deduplication, upsert handling, segmentation, **chunking**, metadata enrichment, and batch publishing for **semantic retrieval**.
- Developed a reusable **MCP runtime harness** combining **IRIS semantic search**, **scoped retrieval**, **tool execution**, prompt assembly, source-ledger tracking, and **grounded response generation** for credit-oversight users.
- Partnered with **cross-functional product, data, and AI teams** to engineer an **agent context harness** managing session state, selected documents, retrieved evidence, policy context, user intent, and **controlled LLM input assembly**.
- Strengthened **operational excellence, observability, and governance** across the **agent runtime** through retrieval logs, execution traces, citation mapping, **guardrails**, cache behavior, and **audit-friendly response tracking**.

Skechers | Software Engineer **Oct 2020 - Mar 2023**

Impact: Delivered **15+ Python APIs** and event-driven services, maintained **99.9% uptime**, reduced costs by **22%**, migrated **35M+ records**, and cut incident resolution time by **40%**.

- Developed **15+ Python/Pydantic REST APIs** on microservices-style, event-driven **AWS Lambda/Step Functions**, sustaining **99.9% uptime** and cutting costs **22%**.
- Migrated **35M+ SFMC records** into **MongoDB** and improved query performance by **50%** through indexing, caching, and data-access optimization.
- Integrated **SessionM**, SFMC, MongoDB, and **AWS serverless services** for loyalty/marketing workflows and **sub-500ms transactions**.
- Applied **event-driven API architecture** across Lambda, **Step Functions**, validation, orchestration, and data-access layers.
- Improved **production resilience** with CloudWatch/Datadog alerts, **API failure tracing**, **retry handling**, and SLA-focused incident feedback.

SOCH Inc. | Full-Stack Developer - Richmond, VA **Jan 2020 - Sep 2020**

- Built a **Spring Boot** ticketing system with **REST microservices** and full-stack workflow screens for government clients.
- Used **Java ExecutorService/CompletableFuture** for async ticket updates, notifications, and background job handling.
- Managed **concurrency** with synchronized services, **thread-safe collections**, timeouts, and Spring Boot exception handling.

FOUNDER-LED 0-TO-1 TECHNICAL PROJECT

OmInsights - Founder-Led Applied AI Platform Build **Jan 2024 - Present**

Impact: Built a **6-component AI platform** with **tenant-scoped APIs**, agent/context orchestration, MCP tool execution, async file processing, and **source-backed intelligence** with tenant isolation, citations, and traceability.

- Architected **Angular UI, FastAPI Business Service, Context Engine, MCP Server**, File Processing worker, and AWS deployment paths with CI/CD, environment promotion, secrets/config handling, and tenant-aware operations.
- Built an **AI evaluation layer** with groundedness, relevance, citation, and policy checks, **LLM-as-a-judge scoring**, and regression tracking.
- Designed **Context Engine flows** for **chat/session memory**, selected context, KB injection, prompt assembly, and **LLM/context harness design** for multi-turn agent orchestration, policy-aware grounding, and session continuity.
- Implemented **MCP/API runtime patterns** and a shared **LLM gateway across platform layers**, centralizing model/provider routing, request policies, retries/fallbacks, tool access, usage logging, and execution traces.
- Created **single-founder execution playbooks** across each platform layer for architecture decisions, implementation, deployment, testing, observability, and recovery, standardizing repeatable delivery and accelerating execution.

TECHNICAL SKILLS

AI Platform & Agent Systems: RAG, embeddings, **vector search**, LangChain/LangGraph, LlamaIndex/LlamaParse, **MCP/FastMCP**, OpenAI APIs, **Amazon Bedrock**, **LLM gateways**, **LLM-as-a-judge evaluation**, **guardrails**, LLMOps, semantic caching, context orchestration

Backend: Java 17, **Python**, Scala, **FastAPI**, Pydantic, **Spring Boot**, **REST APIs**, JWT, microservices, Java concurrency, TypeScript/Angular

Cloud & Data: AWS, **GCP**, Lambda, ECS/S3/SQS, **Step Functions**, API Gateway/EventBridge, Docker, CI/CD, MongoDB, Redis, MemSQL, Kafka, **Flink**/**Airflow**

Retrieval & Storage: OpenSearch Vector DB, MongoDB, hybrid search, metadata filtering/reranking, chunking, **source ledgers**, searchable publishing, Elasticsearch

Observability: Datadog, **CloudWatch**, Kibana, X-Ray, ELK Stack, **structured logging**, SLA alerts, failure tracing

EDUCATION

State University of New York, New Paltz | Master of Science, Computer Science | CGPA: 3.6/4.0 May 2020

JNTUK | Bachelor of Technology | CGPA: 8.5/10 2018